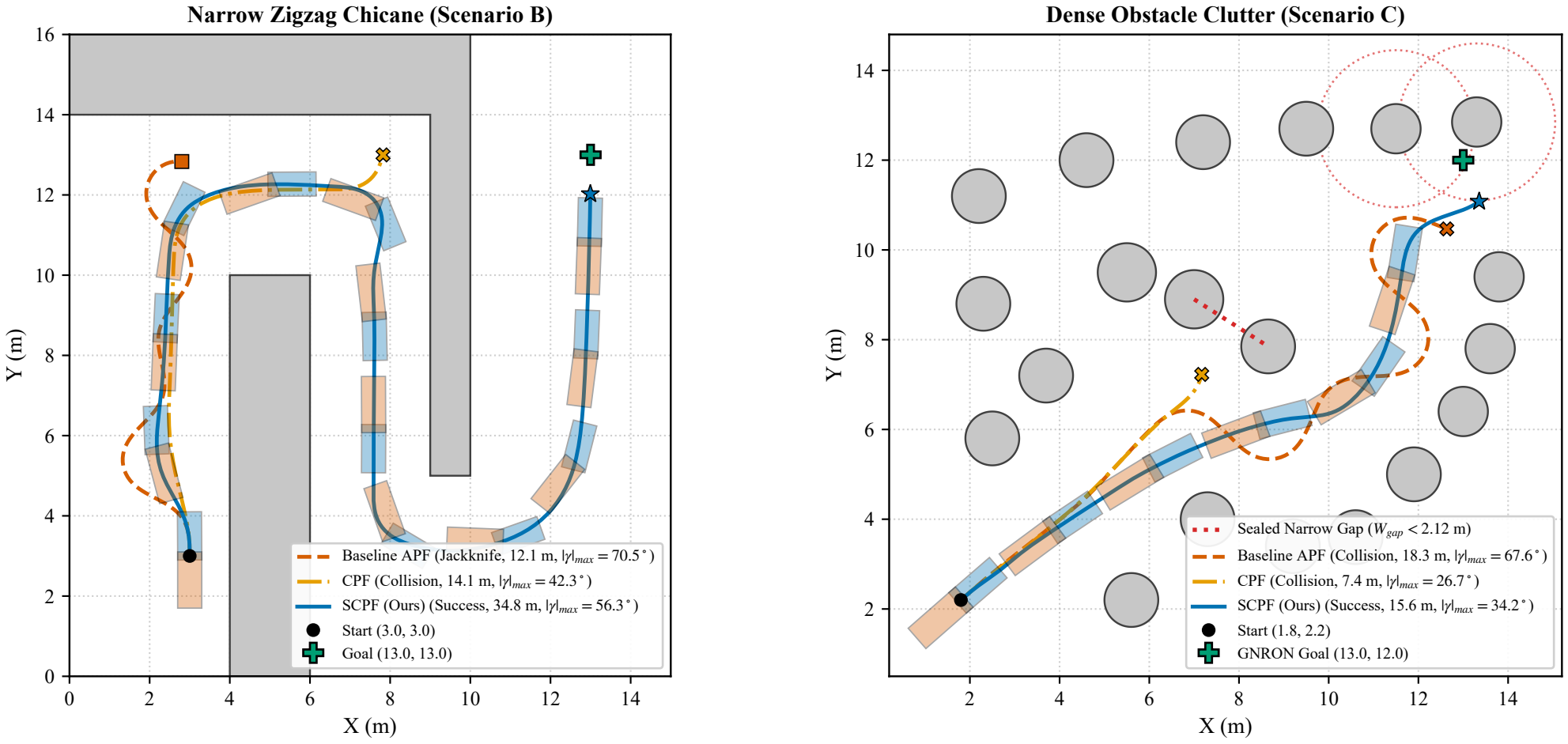

Narrow Zigzag Chicane (Scenario B)
Y (m)
X (m)
Baseline APF (Jackknife, 12.1 m, $|\gamma|_{max} = 70.5^\circ$)
CPF (Collision, 14.1 m, $|\gamma|_{max} = 42.3^\circ$)
SCPF (Ours) (Success, 34.8 m, $|\gamma|_{max} = 56.3^\circ$)
Start (3.0, 3.0)
Goal (13.0, 13.0)
Dense Obstacle Clutter (Scenario C)
Y (m)
X (m)
Sealed Narrow Gap ($W_{gap} < 2.12$ m)
Baseline APF (Collision, 18.3 m, $|\gamma|_{max} = 67.6^\circ$)
CPF (Collision, 7.4 m, $|\gamma|_{max} = 26.7^\circ$)
SCPF (Ours) (Success, 15.6 m, $|\gamma|_{max} = 34.2^\circ$)
Start (1.8, 2.2)
GNRON Goal (13.0, 12.0)